The Operational, Financial, and Business Benefits of Permissionless Blockchains over Permissioned Networks for Users and Regulators

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Abstract:

Growing adoption of cryptocurrencies, stablecoins, and other tokenized assets raises the important question of which type of platform such assets should rely on: permissionless platforms such as Bitcoin and Ethereum where participation in consensus is open to all, or enterprise networks where participation is permissioned and controlled by a gatekeeper.

This paper describes the technical and practical differences between the two types of networks. It explains why permissionless systems offer desirable properties such as resilience, minimized counterparty risk, and guaranteed settlement, and lays out the mechanical process by which the reintroduction of a gatekeeper forfeits those benefits. It argues that permissioned systems are not blockchains and that the assets tracked by them should not be considered tokens.

The paper further compares each design to how legacy payment and capital market systems operate. It argues that permissionless systems are a safer and more innovative upgrade, requiring minimal regulation for responsible adoption. It then demonstrates how permissioned systems are closer to existing systems in operational and financial risk, requiring complex new regulations to maintain stability, protect users, and avoid regulatory arbitrage.

Lastly, the paper contemplates the business implications of the two approaches and shows how enterprise networks are likely to mirror the monopoly power and "too big to fail" dynamics that plague legacy systems, discriminating against smaller entities and startups in favor of large incumbents. It argues for the adoption of permissionless systems as an opportunity to break this paradigm.

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Introduction

The growth of stablecoins and rising excitement over the tokenization of securities has renewed interest in blockchain technology being used to upgrade the financial system. This excitement is warranted. The architecture of most contemporary payment and clearing systems predate the internet and is overdue a digital redesign. Done properly, tokenization can increase financial inclusion, improve capital formation, and reduce risk. But the type of blockchain a token lives on matters, and not all platforms claiming to be a blockchain deliver the associated benefits.

Some blockchains embrace the open and decentralized structure that was originally intended by its inventors, delivering a more innovative yet also safer type of financial infrastructure. Others attempt a compromise that yields the worst of all worlds: infrastructure that is not as safe as what we use today and not as innovative as a properly designed blockchain. The most important distinction is the system's consensus mechanism, the complicated answer to the simple question of who runs and controls the network. Now more than ever, legislators, regulators, and users need to understand the difference between the two approaches. This paper will explore them operationally and financially. It will also make principles-based recommendations on how regulators and users should approach both.

The best blockchain systems (for safety and efficiency) are the ones that aim to maintain decentralization and remain neutral. Neutrality is a highly desirable property for financial infrastructure, for two reasons. The first is that it reduces risk. Infrastructure that operates according to the rules of a protocol and has no centralized points of failure is more reliable than infrastructure that is operated by a handful of players on a discretionary basis. This is particularly true if the rules set forth by the protocol are transparent and difficult to change. The second is that neutrality prevents entities that already rely on the system—and likely govern it—from weaponizing access to deter competition. A more competitive financial system leads to more innovation and lower costs. It also deters Too Big to Fail dynamics.

Today, many legacy domestic and virtually all cross-border settlement systems are owned and operated by large incumbents such as banks, brokers, and card networks. While designed this way for understandable reasons of history, operational efficiency, and compliance, such severe concentration of ownership and governance has downsides. It makes the infrastructure more brittle. A payment system operated by just five players is more vulnerable to mismanagement or cyber attack than one operated by five hundred. Concentration also leads to stagnation and higher costs. Incumbents placed in a position of power due to strong network effects are more focused on maximizing revenues from existing services than investing in new ones. They can perpetually raise the cost of participation or erect other barriers that harm smaller players, particularly startups.

Finance is one of the few industries where small actors are dependent on their larger competitors to serve their own customers, and startups that want to disrupt incumbents effectively require their permission to try, by way of needing permission to use crucial infrastructure. So a community bank that is already bleeding deposits to the Too Big to Fail banks has to rely on a clearinghouse governed

by them to settle payments. And a digitally-native broker that wants to clear its own trades has to fight an uphill battle to integrate with the archaic infrastructure used by the larger firms that dominate securities clearing.

These dynamics are one reason why financial services has lagged behind other industries in terms of digitization. The technology to settle payments quickly or trade stocks continuously has been around for years—markets don't have to close at 4pm. The problem is that the incumbents who control the infrastructure often profit from the inefficiency and the startups offering greater efficiency are locked out of the citadel. They are seldom officially barred from entering, but obscure rules, opaque bilateral agreements, and glacial onboarding schedules achieve a similar effect.

Open and neutral blockchain systems that run on code and cryptography are an opportunity to revisit this approach to building financial infrastructure. Unlike the systems that presently underpin payments and capital markets, public blockchains are not owned by anyone, but can securely be used by everyone. There is no gatekeeper who controls participation. Different functions are performed by different entities, but all who follow the rules of a transparent protocol are welcome. In this regard, blockchain systems are closer to the open protocols commonly found in telecommunications (like TCP/IP) than the closed systems that dominate payments and capital markets.

But unlike the data-transfer protocols that underpin the internet, blockchains are stateful[1] and offer settlement guarantees. Security is provided by a combination of cryptography and incentives, and code does the work traditionally assigned to people. These design elements make public blockchains resilient, from an infrastructure perspective, and predictable, from the user's point of view. Despite the cryptocurrency industry's reputation as being unregulated, some blockchain platforms have the potential to be our most reliable financial infrastructure yet, improving settlement and diminishing counterparty risk. Not because they are decentralized, but because decentralization removes centralized points of failure and makes systems more antifragile.[2]

How a blockchain network is designed and operated matters a great deal. There are many elements to consider, such as consensus mechanism, type of cryptography, and governance. Not every possible combination represents an improvement over the status quo, and there is such a thing as a bad blockchain.

This paper focuses on the most important design element for any blockchain: the question of whether participation in the consensus mechanism used to process transactions is open to all, and thus *permissionless*, or managed by a gatekeeper and restricted to some, which makes it *permissioned*. It will first explain this difference in simple terms. It will then lay out the technical foundation for why permissionlessness is the one design element from which all the other desirable features follow, and explain how a permissioned network cannot offer the desirable properties generally associated with blockchain. Lastly, the paper will explain how failure by regulators to

[1] In computer science, a system is stateful if it has a memory of previous events. See: https://en.wikipedia.org/wiki/State_(computer_science)

[2] The property of becoming more resilient from facing stress, shock, or attack. As popularized by Nassim Taleb

distinguish between these two types of platforms can lead to a riskier financial system and enable regulatory arbitrage.

Today, some practitioners consider the permissioned approach superior to the permissionless one, particularly for regulated activities. They present a gatekept system that still offers certain blockchain-like properties as the best of all worlds: the familiarity of concentrated power combined with new features such as *tokenization* and *smart contracts*.

This view is misguided, as the rest of this paper makes clear. Tokens and smart contracts are not features inherent to any platform that uses cryptography and calls itself a blockchain—in the same way that "final settlement" is not inherent to every database that calls itself a clearinghouse. Tokens are unique assets that can only exist on certain kinds of decentralized systems. Ditto conditional transactions executed inside a decentralized virtual machine, more commonly referred to as smart contracts. There are no tokens or smart contracts on a permissioned network, for the same reason that there are no tokens or smart contracts on a shared spreadsheet in Google Docs.[3]

Assuming that permissioned networks offer the same benefits as permissionless ones is dangerous, for both market participants and regulators. Participants who decide to use gatekept systems face heightened counterparty and settlement risk. They also risk becoming a permanent victim of vendor lock-in and monopoly pricing. Regulators who don't distinguish between the two types of systems risk normalizing infrastructure with fewer settlement guarantees than what is already in place today. They also risk opening the door to regulatory arbitrage.

The centralized settlement systems that power today's systems for payments and capital markets have risks, but they also have controls. Internal rules and external regulations mitigate network outages, erroneous processing, and cyberattack. These controls are based on decades of operating experience. Permissionless blockchains like Bitcoin and Ethereum can operate reliably without these controls because they bring their own: A transparent protocol that adjudicates every interaction, distribution of responsibilities, redundancy, and transaction transparency.

Thus the reliability of a properly designed permissionless blockchain is endogenous to the system. What users (and regulators) see is exactly what they get.

Permissioned networks are different. There is a gatekeeper whose position is enshrined. Neither consensus nor cryptography can constrain its behavior—its power sits upstream of the protocol. This feature allows it to exercise arbitrary power, injecting uncertainty. The gatekeeper can censor specific users, expel disliked validators, or undo settlement. It can also unilaterally change the protocol. Unlike a permissionless system, where the collective will of the validators is king, permissioned networks have a kingmaker, one that can only be constrained by off-platform contracts or regulations.

[3] On a technical level, a permissioned system and Google Sheets are both programmable ledgers that can be used to track the ownership of assets. Both can be distributed among different users, operate on a consensus mechanism, and use cryptography. Neither offers a reliable settlement guarantee.

Thus the reliability of every permissioned platform is exogenous to the system. This is why they require more regulation than their permissionless counterparts; they are less automated and more unpredictable. In this regard, permissioned platforms are closer *in output* to today's legacy infrastructure. But their inputs are different enough to warrant new regulations to make them safe enough to use. What's more, every permissioned platform is different from every other. To paraphrase Tolstoy, decentralized systems are all alike, but every permissioned system is centralized in its own way. Permissioned systems require bespoke regulations to be viable.

Composing new regulations is tricky, for any kind of infrastructure. There may be gaps that only get exposed later, possibly during a crisis. This risk is worth taking for permissionless blockchains because they offer unique properties, such as tokens and smart contracts. Permissioned networks represent a similar risk without the associated reward.

1. Permissioned vs. Permissionless

The most important design question for any distributed system is who gets to participate in consensus. For financial activity, the permitted parties get a say over which transactions are initially entered into a shared ledger and eventually finalized. Distributed systems achieve this via rounds of voting among participants.

For decades, all of the important research on distributed consensus focused only on the permissioned approach, one where some authority constrained the voter set.[4][5] This was for two reasons. First, the earliest distributed systems were hardware-based systems where a single entity controlled every component anyway.[6] Distribution was mostly seen as a way of achieving redundancy, so it was fine to rely on a trusted third party to hand out read and write privileges.
Second, it was generally believed the permissionless approach could not be secure in any adversarial setting where user identities were not known, like the internet.[7]

Bitcoin proved this assumption wrong using a combination of cryptography and incentives.[8] It was the first permissionless system to reach meaningful scale. In the Bitcoin system, participants don't need permission to write to the ledger (as a miner) or read from it (as a node). Lack of permissioning means users can remain anonymous and are free to come and go. But it also makes securing such a system with any kind of voting difficult because one participant could always pretend to be many, effectively "stuffing the ballot." In industry parlance, this is known as a Sybil Attack.

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[4] Leslie Lamport, Robert Shostak, and Marshall Pease. 1982. The Byzantine Generals Problem. *ACM Trans. Program. Lang. Syst.* 4, 3 (July 1982), 382–401. https://doi.org/10.1145/357172.357176

[5] Miguel Castro and Barbara Liskov. 1999. Practical Byzantine Fault Tolerance. In *Proceedings of the Third Symposium on Operating Systems Design and Implementation (OSDI '99)*. USENIX Association, New Orleans, LA, USA, 173–186.

[6] J. H. Wensley *et al.*, "SIFT: Design and analysis of a fault-tolerant computer for aircraft control," *Proceedings of the IEEE*, vol. 66, no. 10, pp. 1240–1255, Oct. 1978.

[7] J. R. Douceur, "The Sybil attack," in *Peer-to-Peer Systems: First Int. Workshop (IPTPS 2002)*, Cambridge, MA, USA, 2002, pp. 251–260.

[8] Satoshi Nakamoto. 2008. Bitcoin: A Peer-to-Peer Electronic Cash System

As described by Douceur in the paper that popularized the phrase:

"..if a single faulty entity can present multiple identities, it can control a substantial fraction of the system, thereby undermining this redundancy. One approach to preventing these Sybil attacks is to have a trusted agency certify identities"

The "trusted agency" is the aforementioned gatekeeper favored by permissioned systems. Bitcoin's approach to Sybil resistance was different. It deployed a mechanism called Proof of Work (PoW) to introduce a cost to voting. As explained by Nakamoto: "If the majority [vote] were based on one-IP-address-one-vote, it could be subverted by anyone able to allocate many IPs. Proof-of-work is essentially one-CPU-one-vote." Proof of Work makes participation in consensus open, but costly. Miners expend resources on compute to participate. Newer blockchains such as Ethereum achieve Sybil resistance via Proof of Stake (PoS), a purely financial mechanism.

A Sybil-resistant consensus mechanism is an important ingredient for securing a permissionless system, but is insufficient on its own. Platforms like Bitcoin and Ethereum require other components to remain decentralized but secure. The most important of these components are:

I) *A native currency:* The miners or validators who participate in PoW or PoS are for-profit mercenaries. They must be compensated for the resources they expend to participate in consensus. This is done via a native currency. Participants usually earn a combination of inflationary rewards issued by the protocol and transaction fees paid by users. Even though some blockchains transact other currencies (such as stablecoins), the native coin is always preferred for consensus because it has the least counterparty risk.

II) *Independent nodes:* Someone needs to double check the work of miners and validators to confirm they've followed the rules of the protocol before they can be compensated. This function is performed by nodes, entities who store their own copy of the ledger and only accept valid transactions to add to it. Nodes are also the channel by which anyone can query the ledger.

III) *Effective Immutability of the record:* Relying on independent (yet anonymous) nodes to maintain copies of the ledger introduces a new problem: how does one node know that another has not tampered with past entries, possibly to inflate its own balances? A distributed ledger is only useful if there is a canonical record of the transaction history.

Bitcoin and other permissionless systems rely on a type of cryptography known as hash functions to create an easily verifiable digest of that history. Hashing allows the honest nodes to quickly detect anyone who has tampered with that record and ignore them. Blockchains are sometimes described as being "immutable" or "append-only", but this is technically incorrect. Every node has the power to alter the data in its own copy of the ledger. Hashing only makes the overall system *effectively* immutable. One official record, maintained by thousands of participants.

IV) *Cryptographic identity*: Every ledger system requires an identity scheme to authenticate users. Whereas conventional systems rely on the users' legal identity (as verified via some kind of KYC

process) permissionless systems use cryptographic identity. The ledger associates each balance with a public address, and transaction instructions are signed with a corresponding private key. This allows users to authenticate themselves while remaining pseudonymous, diminishing the risk of discrimination and censorship.

Table 1: Contrasting key design elements

Dimension	Permissionless Blockchain	Permissioned Network
Consensus participation	Open to anyone	Restricted by a gatekeeper
Sybil resistance	Economic cost to vote (PoW, PoS)	ID certification via trusted-third party
Locus of power	The protocol + collective will of miners/validators & nodes	The gatekeeper + off-platform contractual agreements
Native currency	Required for security	Not required
Independent nodes	Permitted; cannot be revoked	Usually not permitted; If permitted can later be revoked
Identity model	Cryptographic	Legal (via KYC)

Academically, permissioned and permissionless systems have a lot in common. Both follow a consensus mechanism to update a ledger. Both can use cryptography, although permissioned systems don't have to. Both are marketed as useful infrastructure that allows parties who don't trust each other to agree on certain facts.[9] It is therefore understandable why certain market participants would prefer the permissioned approach—decentralization has its drawbacks.

Permissionless systems are more complicated and difficult to scale. They rely on a native coin that is difficult to value and has historically been volatile. Their open nature leads to an ever-present fear that nefarious actors will either attack them—or worse—use them to commit crimes. Cryptocurrencies like Bitcoin have been synonymous with illicit activity from the start. These claims are always exaggerated,[10] especially when compared to the significantly higher amount of illicit flows in the traditional financial system.[11] But the fear is nevertheless there. As are questions about compatibility with financial integrity laws.[12]

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[9] Karl Wüst and Arthur Gervais. 2018. Do you Need a Blockchain? In *2018 Crypto Valley Conference on Blockchain Technology (CVCBT)*. IEEE, Zug, Switzerland, 45–54. https://doi.org/10.1109/CVCBT.2018.00011

[10] https://www.chainalysis.com/blog/2026-crypto-crime-report-introduction/

[11] https://verafin.com/nasdaq-verafin-global-financial-crime-report/

[12] Rettig, Rebecca and Malekan, Omid and Mosier, Michael, *The Compatibility of Permissionless Networks and Financial Integrity: A Practical Guide for Financial Institutions* (August 23, 2026). Available at SSRN: https://ssrn.com/abstract=7343938 or http://dx.doi.org/10.2139/ssrn.7343938

So why not take some of the components of a permissionless blockchain and deploy them exclusively among a restricted group of known counterparties? Such a solution would theoretically be more scalable and can offer additional benefits, such as privacy. Outside users could still benefit from the existence of it by way of the intermediaries who are allowed to participate in consensus. Said intermediaries could now conduct KYC, eliminating all undesirables. On paper, such a system could work.

But financial infrastructure does not exist in an abstract academic setting. It is a kind of plumbing that must operate under adverse conditions in the real world, catering to different users with different needs. Like any other kind of infrastructure, it can suffer from operational issues and shut down. It can act as a honeypot for cyberattack, particularly in the age of AI. If the platform fails operationally, it can cause disruptions in settlement. If it succeeds, it can become a centralized source of counterparty risk. If it fails economically, it can be abandoned and cause a market disruption. If it succeeds economically, it can turn on its participants and exert monopoly power. It can also be coopted by large incumbents to gain an anticompetitive edge over smaller rivals or startups hoping to disrupt them. It is under these real-world conditions where permissioned systems lose most of their appeal.

2. Benefits of permissionless systems for financial activity

The benefits of permissionless blockchains as financial infrastructure fall into three categories: operational, economic, and business. Permissionless blockchains run continuously and always do what they are supposed to do. From the user's perspective, they offer resilience, minimal counterparty risk, and guaranteed settlement. These features are interrelated but will be analyzed separately. Put together, they combine to form a further benefit, one lacking in today's privately owned payment and capital market infrastructure. Permissionless systems can't be captured, even if they become foundational. They can't change their economics to benefit operators at the expense of users. They can't exercise monopoly power, even if they become a monopoly.

Understanding their operational benefits makes it easier to appreciate the financial and business ones. In distributed systems, the two most important operational properties are liveness and safety.

Liveness

Liveness is the property that something good eventually happens. For financial services, it is a measure of reliability. Are transactions processed in an orderly and reliable fashion, so much so that participants can assume the activity they rely on happens in a perfectly predictable fashion? Liveness is crucial for activities such as real-time payments, leveraged trading, or foreign exchange. An individual hoping to complete a real-estate sale would prefer the payment settle instantly, before the title is transferred. A prime broker executing a margin call would prefer the sale go through instantly, lest the client enter into negative equity. A company executing a foreign exchange

transaction would prefer both legs to execute simultaneously, lest it becomes a victim of Herstatt risk.[13] Note that the "eventual" part of the academic definition of liveness cannot be stretched too far. The best financial infrastructure is one that processes such transactions reliably *and* instantly.

Permissionless blockchains provide strong liveness guarantees. These promises originate from the fact that the system is nothing more than a transparent protocol being executed by independent parties, each for its own business interests. Since the system is permissionless, anyone can be a validator helping process transactions, a node double-checking said validators and storing the results for its own uses (or to share with others), or a user submitting transactions.

In Bitcoin, mining can be concentrated (due to the economies of scale to Proof of Work) but the largest miners are nevertheless pools of smaller participants working together. The exact geographic distribution of miners is unknown due to the pseudonymous structure, but it is estimated that 37% of mining occurs in the U.S., 30% is split between Russia and China, and the rest dispersed across Europe, Asia, and South America.[14] Global node distribution is even more dispersed, both geographically[15] and in terms of physical infrastructure.[16] Note that while mining is an industrial activity, a node can be run on consumer hardware.[17]

Table 2: Reachable Bitcoin nodes by origin

	Nodes
All reachable nodes	26,703
Unknown origin	13,186
United States	3,371
Germany	1,536
Australia	207
Singapore	155

Source: BTC Nodes (https://btcnodes.io/) accessed September 7, 2026. Country entries shown are a sample and not consecutive.

In Ethereum, staking distribution is much more dispersed, geographically and institutionally. There are over one million validators globally[18]. Many are run by the same institutions, including ones that

[13] The settlement risk that arises when two legs of an F/X transaction are processed separately and with a delay. See Galati, G. (2002, December). Settlement risk in foreign exchange markets and CLS Bank. *BIS Quarterly Review*, 55–65. https://www.bis.org/publ/qtrpdf/r_qt0212f.pdf
[14] Source: https://data.hashrateindex.com/network-data/global-hashrate-heatmap
[15] See https://btcnodes.io/nodes/live-map/
[16] See https://bitref.com/nodes/map/
[17] See https://bennet.org/learn/building-a-bitcoin-node-with-raspberry-pi/
[18] See https://dune.com/hildobby/eth2-staking

offer staking-as-a-service. But these entities are nevertheless acting on behalf of millions of unique stakers and must keep client economic interests in mind while participating in consensus. The largest such provider Lido—currently accounting for a fifth of all staked ETH—is itself decentralized, distributing the staked ETH to 35 different operators.[19] Ethereum staking can be done with a home PC and reliable internet.[20] Ethereum also boasts over 20,000 nodes who maintain a copy of the ledger.[21] They sit geographically in such dispersed locations as the Arctic Circle, Hawaii, the Canary Islands, and New Zealand.

Table 3: Ethereum stake distribution

	ETH staked	Share
All staked ETH	42,600,000	100%
Lido	9,108,353	21.4%
Binance	3,273,297	7.7%
ether.fi	1,667,239	3.9%
Kraken	1,645,455	3.8%
Coinbase	1,594,039	3.7%
All other stakers	25,311,617	59.4%

Source: Dune Analytics (https://dune.com/hildobby/eth2-staking) accessed September 7, 2026. Total staked ETH as reported by the dashboard; "All other stakers" category consists of smaller staking-as-a-service providers and solo stakers.

Permissionless blockchains that are designed to be more performant than Ethereum, such as Solana, have higher hardware and bandwidth requirements and are more concentrated in terms of validation and independent nodes[22]. But they are still far more diversified operationally than our legacy financial solutions—or any permissioned system—for the simple reason that they are open. Not requiring permission (or a license) to help run a system will always invite a more geographically distributed, operationally independent, and economically diversified group of participants to contribute. Every individual, business, or government who has any connection to Bitcoin or Ethereum can run its own node cheaply, and there are service providers who will help them do so.

What's more, participants in any permissionless system are less likely to have technical or operational co-dependencies than a centralized or permissioned counterpart. They are more likely to use different cloud providers or run the infrastructure on prem. Even if a substantial minority

[19] See https://operators.lido.fi/module/1
[20] See https://ethereum.org/staking/solo/
[21] See https://enrscout.ethnodeops.xyz/?network=mainnet
[22] See https://solanabeach.io/

rely on a single cloud provider, the fact that at least some validators rely on a different cloud provider or run their own hardware aids liveness. Permissionless validators are also more likely to use different internet service providers or sources of electricity. In the case of Ethereum, they are more likely to use different client software. Unlike any legacy system or permissioned network, Ethereum is multi-client, meaning participants acting as validators or nodes use different software made by independent teams to interact with the network.[23]

Table 4: Ethereum execution layer clients

Client	Nodes	Share
Geth	3,130	39.15%
Nethermind	2,654	33.20%
Reth	1,258	15.73%
Besu	563	7.04%
Erigon	141	1.76%

Source: Ethernodes (https://ethernodes.org) accessed September 7, 2026.

Put together, these elements help blockchains like Bitcoin and Ethereum provide the strongest liveness guarantees of any electronic financial infrastructure in operation today. Both have operated around the clock—including during nights, weekends, and holidays—for the past decade without an outage. Notably, Bitcoin's liveness is stochastic. The protocol targets a new block every ten minutes but the next one can come in seconds or in an hour. Proof-of-Stake systems are more predictable. Ethereum generates new blocks every 12 seconds, and there are proposals to shorten that interval.[24]

Ultimately, the reliability of these systems comes from their malleability. The operation of key infrastructure is so diffused that it can withstand shocks and black swan events. They are not overly dependent on any one company, country, internet-service provider, software maker, or power source. Other than being permissionless, Bitcoin and Ethereum further incentivise those who contribute to liveness and punish those who don't. Bitcoin pays each miner a combination of inflationary rewards and transaction fees to process the latest batch of transactions. Ethereum does the same for validators, and goes one step further by confiscating the stake of validators who don't contribute.

To be clear, not all permissionless systems are this resilient, and some blockchains optimize for other features, such as throughput or low latency. But the ceiling of reliability for any decentralized

[23] See https://ethernodes.org/
[24] See https://eips.ethereum.org/EIPS/eip-8198

and permissionless system is always higher than that of any centralized infrastructure or permissioned platform. The users of permissionless systems still have to pay sufficient fees for timely processing and contend with network congestion, but the rules of engagement are clear, and difficult to change.

A centralized system or permissioned network can never provide liveness guarantees that are this strong. If a single entity is in charge of updating the ledger—as happens to be the case for most modern payment and clearing systems—it can always go down, suffer an outage, experience a bug, or fall victim to an attack. The permissioning authority in any permissioned system is similarly vulnerable, it is a single point of weakness. Redundancy and backup systems mitigate these risks, but they can never reduce them to the extent a permissionless architecture does.

Lastly, permissionless systems enable peer-to-peer transactions. Lack of intermediation means the main counterparty users face is the infrastructure itself. The more reliable the infrastructure, the lower the counterparty risk everyone faces. For any centralized clearinghouse—like the ones that underpin virtually all payment and capital market activity today—users have to worry about the reliability of the clearinghouse *and* the intermediary they are forced to use to access it. This dynamic creates a compounding problem. The Federal Reserve doesn't publicly disclose outage data, but there have been reports of multiple instances of degraded service[25] or total outage over the past 5 years.[26] This number needs to be multiplied by the immeasurable number of times the banks who rely on the Fed for wholesale payments have had their own internal issues to get a sense of how frequently a payer had to face a disruption. Disintermediation makes systems more resilient because the chain of activity has fewer links that can break.

Safety

Safety is the property of nothing bad ever happening. In layman's terms, it can be understood as a transaction never being reversed after it is written into the ledger. Financially, safety is a measure of final settlement. Whereas liveness is a measure of how quickly and reliably transactions are processed, safety is a measure of what happens afterwards.

Permissionless blockchains provide strong safety guarantees thanks to the same robustness features discussed above. The more diversified the participants who contribute to running the ledger, the less likely that any one has the power to reverse settlement. Diversification also diminishes the risk of collusion. Centralized solutions struggle to provide similar guarantees. Most payment systems and capital-market clearinghouses rely on a single entity to maintain the ledger. Some, such as central-bank-run RTGS systems, offer strong settlement guarantees.[27] But these promises are only as good as the off-platform contractual agreements that exist between operator and user, and the regulations that oversee those agreements. Importantly, legal agreements can't

[25] See https://statusgator.com/services/federal-reserve/outage-history

[26] See https://www.wsj.com/finance/regulation/federal-reserve-reports-several-business-lines-disrupted-by-operational-error-11614194912?mod=article_inline

[27] Real-time gross-settlement systems are designed to process every transaction instantly and in irreversible fashion. They are used in payments or securities transfers.

prevent the operator from experiencing a bug or falling victim to a cyberattack. Settlement provided by a single authority is always vulnerable to mechanical malfunction.

Importantly, centralized systems seldom run continuously. They often shut down on nights and weekends to perform maintenance. They are also always restricted to an exclusive set of large intermediaries. Market participants—be they individuals or large institutions—who want to benefit from guaranteed settlement can only access such systems via an intermediary such as a bank or broker. This requirement makes them vulnerable to all the things that can go wrong with any intermediary, such as errors, outages, cyberattack, or even bankruptcy. It also makes users vulnerable to the things that can go wrong *in their relationship* with an intermediary, such as high fees, discrimination, censorship, or debanking. Meanwhile, the safety guarantees of any permissionless system are available to anyone, anywhere, at all times.

Permissionless blockchains offer strong safety guarantees that translate directly to final settlement. These guarantees flow from a combination of the technical components and the use of financial incentives. Hashing makes it easy for nodes to continuously agree (then re-agree) on the canonical history of transactions, rendering the ledger effectively immutable. Transparency makes it easy for any user to also be a node operator and confirm their transaction has not been reversed.

In Proof of Work systems, miners who attempt to alter the history of transactions require majority compute power to even try. Even then, they face a strong economic incentive to not try. Why amass majority resources to run an infrastructure only to diminish its appeal? Bitcoin miners get paid in Bitcoin and would like to see its value appreciate. The best way for them to contribute to that cause is by preserving the safety of the network. Meanwhile, the cost of even attempting such an attack is astronomical.[28]

In Proof-of-Stake systems, validators who attempt to alter the ledger require majority stake to even try. Acquiring that many coins would make the price skyrocket, creating a strong incentive to not even try. The Ethereum protocol is designed to detect any such attack and confiscate the coins staked by the offender. It is also designed to detect any collusion by multiple validators and to punish them proportionally.

As an added protection, permissionless blockchains such as Bitcoin and Ethereum are designed so that the nodes have the final say over which version of history is valid and which isn't. In the (highly unlikely) event that all of the above mitigations fail, node operators can still collude to ignore an attack. Doing so would require off-platform coordination, but the threat of nodes doing so acts as a final check on malicious miners and validators.

A financial infrastructure that provides such strong safety guarantees significantly reduces settlement risk. Networks such as Ethereum further reduce settlement risk by being omni-asset and programmable. Most of our legacy settlement systems are single asset or even single purpose for an

[28] Nuzzi, Lucas and Waters, Kyle and Andrade, Matias, *Breaking BFT: Quantifying the Cost to Attack Bitcoin and Ethereum* (February 15, 2024). Available at SSRN: https://ssrn.com/abstract=4727999 or http://dx.doi.org/10.2139/ssrn.4727999

asset. For example: Fedwire for dollars, T2S for euros involved in capital market settlement, and Euroclear for European securities. An American investor hoping to purchase European stocks has to rely on multiple intermediaries to complete the transaction, increasing counterparty and settlement risk. There are bespoke solutions meant to address this issue (such as CLS in F/X trading) but they are limited. Complicated financial transactions always straddle multiple systems and clearinghouses, relying on intermediaries such as banks and brokers to facilitate settlement. The problem is compounded in any cross-border transaction. Each intermediary is a link in a chain, and it only takes one weak link to break the chain of final settlement. Settlement risk is an ever-present concern, particularly during crises.

Permissionless systems such as Ethereum reduce this type of settlement risk thanks to tokenization and smart contracts. Any asset can be turned into a token, and the interaction of every token with every other could be facilitated with code. Thus the strong settlement guarantees of a blockchain are multi-dimensional. They apply as much to a complicated investment requiring an F/X swap and securities purchase as they do to a simple payment. This also means market participants face even less counterparty risk than mentioned above, because the infrastructure performs the work of multiple intermediaries, atomically, and with cryptographic guarantees.

Diminishing counterparty and settlement risk in such a fashion has been the holy grail of financial system engineering and regulation throughout history. And even though there are always new centralized solutions meant to offer instant settlement, delivery-vs-payment, or payment-vs-payment, none can do it for every asset, in real-time, and across any border. Nor can they handle both a capital market activity, such as dividend payment, and the resulting payment, in one system. Nor can they handle transactions in a derivative and the underlying it settles to on a single platform.

Put together, the liveness and safety guarantees of a permissionless blockchain, combined with an omni-asset design and programmability, represent an exponential reduction in operational, counterparty, and settlement risk. Academically, tokens and smart contracts on a blockchain platform are just features of a new kind of platform. But when used by real-world participants for payments and capital markets, these features provide for a cheaper, more innovative, and safer financial system. They should be embraced by users and regulators.

Business Benefits

Most of the debate about financial infrastructure is focused on what happens when the system falters, like during an outage or financial crisis. Not enough attention is paid to what happens if the system succeeds, and the disproportionate power success places in the hands of operators over everyone else. Payments and capital markets are prone to strong network effects. Once a clearinghouse, settlement system, or ancillary solution becomes central to economic activity, it is unlikely to be dethroned. What's more, many such systems eventually get their position enshrined by statute or regulations, despite being privately owned. The result is oligopolistic or monopolistic behavior that can turn corrosive.

The concentration of financial infrastructure is not in dispute. Visa and Mastercard account for almost 90% of card purchase volume in the United States.[29] Also in the U.S., exchange-traded equity options, Treasury securities, mortgage-backed securities, and public equities each have a single dedicated clearing agency.[30] Several of these agencies are owned by a single private corporation, the DTCC, whose shares are held by the large intermediaries who rely on it as securities depository and clearing agency. Similar concentration exists in the payment and settlement systems of most countries.

To be fair, concentration in financial infrastructure has its benefits. The less fragmented the overall payment system, the more frictionless the payments that rely on it. Also, a single clearing agency for the same security is easier to supervise and regulate than a complicated network of parallel agencies. Nor does concentration imply that anything nefarious has happened. Payment systems have strong economies of scale, and the unification of securities clearing may be a byproduct of regulations that make it harder for smaller agencies to compete, driving consolidation.[31]

Regardless, concentration of key infrastructure has downsides. Most of our payment and capital market infrastructure is now systemic and too big to fail. This status imposes an unfair burden on taxpayers if a bailout becomes necessary. The card networks are chronically the target of antitrust investigations and merchant lawsuits. Clearinghouses such as the NSCC are also the subject of controversy, particularly during events like the memestock craze.[32]

Arguably the greatest costs of such concentration are the ones that are hard to measure, the ones we cannot see. Payments remain stubbornly expensive, despite the digital revolution. Securities trading happens in milliseconds, but settlement requires an entire day (or longer over weekends and holidays). Community banks have to rely on infrastructure owned and operated by the biggest banks, the same ones who have been siphoning their deposits thanks to their Too Big to Fail status. FinTechs have to ask permission to use infrastructure operated by the incumbents they hope to disrupt.

Interoperability remains lacking, despite all markets being electronic and the world being connected by a single internet. Physical assets sometimes cross borders more easily (and cheaply) than money or securities. The architecture of financial settlement has not changed very much since the dawn of the internet.[33] Neither have the intermediaries who are pivotal to payments or capital markets, due in part to their Too Big to Fail status. The most likely driver of change in market structure is either consolidation or catastrophic failure, not innovation.

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[29] See https://nilsonreport.com/articles/mastercard-and-visa-cards-in-the-us-2025/

[30] See https://www.sec.gov/files/rules/final/2023/34-98959_0.pdf (page 137)

[31] Dan Awrey & Joshua C. Macey, *Open Access, Interoperability, and DTCC's Unexpected Path to Monopoly*, 132 YALE L.J. 96 (2022).

[32] See https://fortune.com/2021/02/02/robinhood-gamestop-restricted-trading-meme-stocks-gme-amc-vlad-tenev-nscc/

[33] Omid Malekan, *Re-Architecting Trust: The Curse of History and the Crypto Cure for Money, Markets, and Platforms* (Pennsauken, NJ: BookBaby, 2022).

Counterfactuals are always difficult to measure, but public permissionless networks now provide an alternative for comparison. Unlike a card network, Ethereum cannot exercise pricing power over individual users. Nor can it give preferential access to a large bank over a community one. Public-permissionless blockchains force miners or validators to treat every user equally, lest they face a penalty. What's more, blockchain fees are only derived from network congestion and the complexity of the transfer (e.g., does it interact with a smart contract?), they do not consider the size of the payment.

In contrast, the card networks post the discriminatory prices they charge different industries on their websites.[34] The same dollar paid to an ecommerce website costs more to transfer than it would to a brick-and-mortar store, even to purchase the same product. The operators of these networks have been accused of anti-competitive behavior by way of undisclosed private arrangements.[35] They also charge fees (either for themselves or on behalf of issuers) that are proportional to the amount being sent—a practice that makes little sense in the digital era. Payments are just ledger entries, and entering $1 into a digital database takes the same computational resources as entering $1 million. Proportional fees on payments are yet another indication of market power. Blockchains like Ethereum and Solana charge the same fee on a $10 payment as they do a $10 million one, across the room or around the world.

Permissionless systems are similarly appealing for capital market settlement. Anyone can issue a tokenized security on a decentralized blockchain, and anyone else can invest in that token. Same for the stablecoin used to purchase tokenized securities. This is what it means for a permissionless network to be *censorship resistant*. The issuers of these tokens need to be regulated in the traditional sense, but the blockchain used for clearing and settlement does not. The protocol ensures open access and orderly execution.

These business benefits have important downstream consequences. The individuals and businesses who may someday come to rely on decentralized infrastructure can expect continued access, regardless of the network effects the platform attains. So can the banks, brokers, and FinTechs who hope to plug into blockchain infrastructure to better serve their customers. The lack of a holdup problem allows them to invest in their business freely.[36] Nor do these firms have to fear the economic benefits of their activity flowing to an incumbent or competitor who happens to be the operator of crucial infrastructure. Permissionless blockchains don't have equity in the traditional sense, and there is no obvious way to capture the value created by their success. The closest proxy is the native coin, and those can be owned by anyone.

Lastly, the censorship resistance of permissionless blockchains, combined with their promise of business continuity, leads to far greater innovation. Blockchains have been home to exciting new primitives such as automated market makers and perpetual swaps. Both innovations are poised to

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[34] See https://usa.visa.com/content/dam/VCOM/download/merchants/interlink-interchange-reimbursement-fees.pdf and https://www.mastercard.com/content/dam/mccom/us/business/documents/merchant-rates-2026-2027-april2026.pdf

[35] https://www.justice.gov/archives/opa/pr/justice-department-sues-visa-monopolizing-debit-markets

[36] See https://en.wikipedia.org/wiki/Hold-up_problem

feature prominently in the capital markets of the future. The fact that incumbent market makers and derivative exchanges have responded to these developments with innovation-killing regulatory proposals[37] and lawsuits[38] confirms their disruptive potential.

Some incumbents have responded to these innovations by embracing permissioned networks and calling them blockchains. This approach is based on a misunderstanding of what makes something a blockchain to begin with and where the benefits stated above actually come from.

3. The shortcomings of permissioned networks

There is a popular fallacy that permissioned protocols such as Hyperledger Fabric or R3 Corda are blockchains in every way other than their restricted validator set. This view is based on a misunderstanding of how the different components of an actual blockchain system come together to provide the operational, financial, and business benefits catalogued in this paper. The re-introduction of a gatekeeper, one whose primary duty is to control participation in consensus, invalidates most of those benefits. Permissioned networks are closer—in both design and output—to today's legacy infrastructure than to public blockchains. At best, their success will lead to an even more fragmented financial system still plagued by censorship and abuse. At worst they will make the system more unstable.

As a simple example of the false equivalence, the ledger of a permissioned network can never be effectively immutable, even if it deploys hash functions in the same manner as a permissionless blockchain. Most permissioned networks don't allow for independent nodes. Even if they did, that option might be eliminated at any time by the gatekeeper because its power sits upstream of the protocol. Copies of a permissioned ledger are thus only kept by authorized validators. They might periodically hash new entries for reasons of data compression and comparison, but that would not prevent the gatekeeper from ordering every validator to revert the state of the ledger, then rehash everything. Validators who don't comply with this order risk being de-permissioned.

Note that this kind of state reversion has also happened in permissionless blockchains with centralized power and insufficient validator diversity. Ethereum executed one in its infancy over a decade ago.[39] Other blockchains have executed one more recently.[40] State reversion is a form of settlement failure. Hashing alone cannot prevent it, on any type of system. Only a diffuse system, with a diverse set of validators and nodes, and a culture of neutrality, can use hash functions to avoid that fate—thus offering strong settlement guarantees. A permissioned system cannot because the gatekeeper can override every other participant's preference. This is why permissioned systems are not blockchains.

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[37] See https://www.sec.gov/files/citadel-securities-120225.pdf
[38] See https://www.reuters.com/legal/government/cme-sues-us-cftc-over-letting-kalshi-coinbase-offer-perpetual-futures-2026-06-18/
[39] See https://www.gemini.com/cryptopedia/the-dao-hack-makerdao
[40] See https://www.theblock.co/news/defi/2025-11-03-berachain-emergency-hardfork-377257 and https://www.sui.io/blog/cetus-incident-response-onchain-community-vote

The presence of a gatekeeper also forfeits the liveness benefits of a blockchain. Permissioned systems always have far fewer validators than permissionless ones. Constraining the set of participants *is* the point, and each validator has to be onboarded individually, possibly via contractual negotiation. A permissioned system is more likely to have a few validators than a few thousand. Its validators are more likely to have other co-dependencies, such as geographic location or cloud provider. Many permissioned systems are industry consortiums. Worst of all, and regardless of the number of active validators, the authority could always trim the group unilaterally. It can also order all validators to halt the network. There is nothing mechanically preventing this, and the proverbial Sword of Damocles hangs over every participant subject to the power of the gatekeeper.

Some permissioned systems place a non-profit foundation or industry consortium in charge of managing consensus, possibly with the hope of being perceived as more decentralized and safe. Unfortunately, this design choice does not mitigate the resilience and settlement concerns discussed above. Who acts as gatekeeper is less important than the fact that there is one. In practice, every permissioned network is a distributed database with a button, and it is the presence of the button that elevates risk, not the legal structure of its operator. Even the most well-intentioned operator could fall victim to a software bug or cyberattack. Indeed, the authority in charge of the button will be an attractive target for hackers and saboteurs. So are the entities authorized to participate in consensus. Many permissioned networks disclose the identities of their validators publicly. It is easier for hackers to target the 10 banks acting as authorized validators on a consortium network than the thousands of independent validators running Ethereum. Simply taking 4 validators offline would be sufficient to halt such a permissioned network.

Some permissioned networks operate inside a single corporation, like a bank. This is an exercise in futility. Having a single entity act as both gatekeeper and node operator defeats the purpose entirely. A modern distributed database would achieve the same outcome with far less complexity and far more throughput. From a regulatory perspective, such a solution should be treated as inferior to using vanilla core banking infrastructure to fulfil the same function. The cryptography and consensus of this network adds technical and operational risk with no additional benefit.

Most permissioned systems are also private.[41] Not only is participation in consensus gatekept, but so is usage of whatever service the network facilitates. These systems are fully censored, unlike permissionless blockchains that let anyone with a private key initiate transactions. But some of the newer permissioned networks that have launched in recent years claim to be public.[42] Validation is restricted, but usage is allegedly not. These networks claim to achieve public usage via the same type of cryptography used by public blockchains like Bitcoin and Ethereum. But in any permissioned setting, this design is unsustainable.

At issue is the liability taken on by the *known* validators. In a permissionless system like Bitcoin, it is unclear which miner will process the next block, even to the miner that eventually wins the right to

[41] Academically, permissioned vs. permissionless refers to participation in consensus, while private vs. public refers to usage

[42] See https://www.circle.com/blog/introducing-arc-an-open-layer-1-blockchain-purpose-built-for-stablecoin-finance

do so. Ethereum picks validators pseudorandomly, and their identities are not known. What's more, the protocols of both blockchains are designed so a transaction foregone by one miner or validator is simply processed by the next. A Bitcoin transaction ignored by an American miner could be processed by a Chinese one, or one with an unknown location. This is what it means for a system to be censorship-resistant. Ethereum is introducing new tools to make this feature even more robust.[43] This mostly-involuntary approach to updating a ledger has the added benefit of protecting validators from plaintiffs and regulators. Participants can't easily be forced into dispute resolution or be tasked the mitigation of illicit activity.

In a permissioned setting, the identity of every validator is known, and every transaction can be tied to a specific participant. The round-robin nature of most permissioned consensus mechanisms means an illicit funds transfer to a terror organization can be ascribed to a single entity, like a bank, acting as proposer. Meanwhile, the fact that the overall validator set is small, and the participants likely have contractual arrangements with each other, means they can easily collude to block such a transaction indefinitely.

Public-permissionless blockchains aren't expected to enforce such frameworks because they can't.[44] Permissioned platforms that claim to be public can, but choose not to. This is unlikely to persist. Allowing permissioned networks to operate in this fashion will result in undesirable regulatory arbitrage. Why should a permissioned platform for capital market activity be exempt from the illicit finance laws that constrain a clearinghouse owned and operated by the same companies? The long-term arc of every permissioned system is censorship, if only to comply with regulations.

Censorship can cut both ways. Whereas the good kind of censorship combats illicit activity, the bad kind enables discrimination. Permissioned systems that restrict usage can discriminate based on race, ethnicity, age, nationality, immigration status, sexuality, or politics. Even if the validators of such a system don't want to discriminate, they can be forced to by the gatekeeper, or a foreign government. Some governments weaponize access to financial services to punish dissidents and oppress minorities, and a permissioned system preserves the mechanism for doing so. Every permissioned system has a "throat to choke", for better and for worse.

Meanwhile, even in societies where discriminatory practices are banned, the validators of a permissioned system could still discriminate for business reasons. They can give preferential treatment to their own clients or refuse to process the transactions of their competitors. Permissioned networks run by industry incumbents, like the biggest banks, could discriminate against smaller players, like community banks, or disruptors, such as neobanks. To be fair, discrimination and censorship are also risks for the centralized infrastructure the financial system relies on today. But these systems are heavily regulated, in a way permissioned systems are not. Thus permissioned systems would need new laws and regulations, only to achieve the status quo as far as fair access is concerned. Permissionless systems cannot discriminate.

[43] See https://forkcast.org/eips/7805/
[44] Rettig, Malekan & Mosier, The Compatibility of Permissionless Networks and Financial Integrity, supra.

The ever-present risk of censorship on any permissioned network also eliminates the strong property rights now associated with actual blockchains. Permissioned systems cannot offer bearer assets like cryptocurrencies; their validators can collude to censor specific owners from transacting their assets or be compelled to by the authority. This is why the assets transacted on any permissioned system are not tokens. Tokens offer strong property rights due to the liveness, safety, and censorship-resistant properties of actual blockchains. Permissioned platforms can forfeit these features with the push of a button—their implied property rights are weak.

Nor can any permissioned system offer smart contracts. Any type of ledger (however centralized) can offer programmability via APIs, and many legacy ledgers do. What makes smart contracts unique are the execution guarantees provided by a decentralized and neutral infrastructure. Smart contracts are conditional transactions guaranteed to satisfy all parties who agree to their terms.[45] Permissioned networks cannot offer such a guarantee. The validators who update the ledger can collude to ignore code, change it, or refuse to process the resulting transaction. There is nothing mechanically preventing this.

Defenders of these systems often argue such violations are highly unlikely, even if they are technically possible. They argue the power of the gatekeeper can be constrained with off-platform contractual agreements. They also point to the reputations of the *known* validators as a further bulwark against undesirable behavior. This argument is flawed for several reasons.

First, contracts and reputation have nothing to do with the *physical* vulnerability that comes from concentrated operation. The existence of a single point of failure, and the reliance on an overall smaller set of validators, makes these networks more vulnerable to operational issues and cyber risk. Second, contractual obligations have not historically impeded the oligopolistic or monopolistic behavior of legacy system operators. If anything, the complexity and opacity of these arrangements have only enhanced their market power and further entrenched their Too Big to Fail status. A permissioned network for payments that is held together by secret contracts among the participating banks is more likely to discriminate against other banks and FinTechs, not less.

[45] Omid Malekan, *The Story of the Blockchain: A Beginner's Guide to the Technology That Nobody Understands* (New York: Triple Smoke Stack, 2018).

Table 5: Practical implications of consensus design

Dimension	Permissionless	Permissioned
Liveness Guarantees	High due to uncapped number of validators & lower likelihood of shared dependencies	Low due to capped validators, higher odds of shared dependency. Gatekeeper can halt network
Cyber risk	Low due to lack of single point of failure, validator diversity, some validators being unknown	High due to single point of failure & smaller number of known validators
Immutability	Effectively immutable due to hash functions deployed by diverse nodes	Gatekeeper can order state reversion regardless of use of hashing
Safety guarantees	Endogenous — coming from protocol design	Exogenous — depend on off-platform contracts
Censorship	Resistant due to validator diversity & pseudonymous usage	Possible due to validator collusion & reliance on KYC
Tokens	Available; Strong property rights emanating from protocol design	Unavailable; Weak property rights contingent on off-platform contracts
Smart contracts	Available; Execution guaranteed by protocol design	Unavailable; Validators can ignore, alter, or refuse to execute code
Pricing	Neutral; Set by congestion and transaction complexity	Discriminatory; Set by gatekeeper and/or validators
Capture risk	Cannot be captured, even if platform becomes foundational	High, particularly if run by industry consortium
Regulatory posture	User protections are endogenous and require minimal regulation	User protections are exogenous and require strict regulation

4. The risk of regulatory arbitrage

To be fair, permissioned networks can deploy unique tools that legacy payment clearinghouses and settlement systems generally do not, such as soft consensus, hash functions, and public-key cryptography. At least in theory, they can also offer unique features such as public access, transparency by way of independent nodes, and more direct programmability of transactions. These tools and features are meant to mirror the ones of a public permissionless blockchain. The challenge for regulators is that the two types of networks are much closer in input than output.

As mentioned earlier, the end-user protections of any permissionless network are endogenous to the system. Regulators can simply inspect the rules of the protocol and the current state of the network to see if the purported protections are there, and sustainable. Permissioned platforms do not provide this kind of line of sight. The discretionary behavior of the gatekeeper and its preferred validators supersede any technical component, and that behavior needs to be controlled and regulated, just as it is in legacy infrastructure for payments and capital markets.

This unfortunate reality creates a challenge for both insiders and regulators. Every centralized system requires a complex combination of internal rules, operating agreements, laws, and regulations to be viable. Just to give one example, public securities clearing and settlement in the U.S. is controlled and regulated via a 440-page "Rules and Procedures" document at the NSCC, an 80-page "Settlement Service Guide" at the DTC, a 180-page "Rules, Bylaws, and Organizational Certificate" at the DTC, and the many relevant pages in the applicable sections of the 1934 Securities Exchange Act and 2010 Dodd Frank Act.[46] It cannot be that the entities currently controlled by the more than 1,000 pages of rules listed above become exempt from them simply by launching a permissioned network for similar activity.

If the risks are equal—which this paper has argued they mostly are—then the rules and regulations must at least be equally robust. Unfortunately, it also cannot be that the process of clearing and settlement is regulated the same way on a new permissioned system as it is at a legacy clearinghouse. Laws passed in the 1930s and rules established in the 1970s are insufficient on a network using consensus and cryptography.

Thus every permissioned system will require complex new internal rules and external regulations to become trustworthy and safe, or to approach the resilience and risk mitigation of a permissionless system. Not imposing new rules and regulations on permissioned systems incentivises regulatory arbitrage. If all it takes for a payment system or clearinghouse to become exempt from the applicable laws and regulations is to introduce performative cryptography, then it's logical to assume most such systems eventually will. Regulation is always a cost, however justified it might be.

It is likely that most of today's proponents of permissioned systems do so more out of ignorance than an outright desire to escape regulation. They actually believe a permissioned system run by a consortium of banks offers the same benefits as Ethereum, so they argue for a light-touch regulatory approach to both. In their defense, they've been misled for years by a dangerous combination of avaricious software companies, uninformed academics, confused journalists, and overeager cryptocurrency investors.

Legislators and regulators don't have the same luxury of *not knowing*. As this paper makes clear, permissioned networks do not provide the same liveness, safety, and business benefits of

[46] See https://www.dtcc.com/~/media/Files/Downloads/legal/rules/nscc_rules.pdf, & https://www.dtcc.com/-/media/Files/Downloads/legal/service-guides/Settlement.pdf & https://www.dtcc.com/~/media/Files/Downloads/legal/rules/dtc_rules.pdf

permissionless blockchains. If left unregulated, both the counterparties who plug into these centralized systems, and the users who rely on them as intermediaries, face greater risk. The overall financial system will become more fragmented and brittle.

Regulating any new type of infrastructure is not easy. But as money and assets increasingly move on chain, regulators will need to make important decisions. They will need to identify what type of infrastructure helps mitigate risk and which can possibly increase it. They will also need to be careful not to open the door to regulatory arbitrage. Lastly, but just as importantly, they will need to clarify that when it comes to financial infrastructure, desire is not execution. In this regard, a *bad* permissionless system can be just as dangerous as any permissioned one.

Conclusion

The opening paragraphs of the Bitcoin white paper lay out the goal of designing financial infrastructure that two parties can use safely without the aid of a trusted-third party. Almost 20 years later, several permissionless networks have demonstrated the ability to do just that. For much of that time, these decentralized systems have been imitated by more centralized alternatives, referred to interchangeably as permissioned networks, enterprise blockchains, or distributed ledger technologies.[47] The biggest difference between the two types of networks is the re-introduction of a gatekeeper to control participation in consensus.

This paper has made the technical case for why permissioned systems have little in common with the actual blockchains that settle trillions of dollars in value annually. But an even simpler critique is the fact that they reinstated the trusted-third party that blockchain was invented to eliminate in the first place. The academic literature on distributed systems has always waved away the risks of relying on such a warden, even after the invention of Bitcoin.[48] This was easy to do when the presumed applications of permissioned systems were either imaginary or non-financial, like the provenance of produce.[49] Making the same mistake is not an option now that permissioned systems are being proposed as a means of updating our infrastructure for payments and capital markets.

Permissionless systems offer desirable features such as resilience, minimal counterparty risk, and guaranteed settlement. They do so using a novel mix of consensus, cryptography, and incentives. Permissioned systems cannot offer similar assurances due to their gatekept architecture. The fallacy of why they might be comparable (or even desirable) starts with the false belief that the open vs. closed nature of distributed consensus is but one of the important building blocks for a useful network. Proponents of permissioned networks argue they are just as useful as public blockchains, just different. This is like arguing cement mix is but one of the useful ingredients in making

[47] Distributed ledger technology, or DLT, is seldom used to refer to actual blockchains that settle real value
[48] Wüst & Gervais, Do You Need a Blockchain? (2018), supra.
[49] See https://www.prnewswire.com/news-releases/albertsons-companies-to-join-blockchain-based-ibm-food-trust-network-to-pilot-increased-transparency-for-romaine-lettuce-300830716.html

concrete, and that eliminating it yields a building material that is still useful, albeit different. In practice, it only yields wet sand.

The other way to understand the benefits of permissionless blockchains over permissioned networks is to consider the simple answers open networks provide to the most important questions asked by users and entrepreneurs. For example, a FinTech or community bank hoping to plug into the most advanced infrastructure to better serve its clients might ask: Who owns this network? Who can use it? Who profits from it? Who benefits from the data generated by it?

For open systems such as Bitcoin, Ethereum, and a few other blockchains, the answers are: no one, everyone, token holders (who can be anyone), and the public. But for closed systems such as permissioned networks featuring nominal amounts of consensus and cryptography, the answer to each question is just as complicated as it would be for any existing payment system or capital market clearinghouse. Whereas a permissionless blockchain is just a system being contributed to by countless volunteers for their own reasons, a permissioned network is an entity, and entities are owned, managed, and monetized. When they fail they put the rest of the economy at risk. If they succeed, they can become Too Big to Fail and exercise monopoly power.

Practitioners looking to use or build on either type of network should be mindful of their distinctions in terms of resilience, counterparty risk, and guaranteed settlement. Regulators hoping to pave the way for responsible adoption need to be aware of these distinctions as well. Whereas one approach yields infrastructure that is autonomous and predictable, the other requires custom rules. Conflating the two can lead to regulatory arbitrage and a riskier financial system.

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